Exploration of the Intervention Effect of Ethanol Combined with N-Methylthiotetrazole (NMTT)-Containing Cephalosporins on Skin Melanin Overload in Hydrophobia pygmaeus

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Abstract

Objective: To explore the intervention effect of ethanol combined with N-methylthiotetrazole (NMTT)-containing cephalosporins on skin melanin overload in Hydrophobia pygmaeus (commonly known as "water ghosts"), and to analyze the physiological abnormalities, behavioral acquisition, and group protest behaviors induced by the drug combination, so as to provide references for pigment regulation and ethological research on unconventional model organisms.

Methods: Thirty-six healthy adult Hydrophobia pygmaeus were selected from the freshwater basins of the middle and lower reaches of the Yangtze River and randomly divided into 4 groups (n=9 each): blank control group (normal saline gavage), ethanol group (4.5% v/v industrial lager beer, 5ml/kg gavage), cephalosporin group (cefoperazone sulbactam, 10mg/kg gavage), and combined group (ethanol + cephalosporin at the same doses). Skin melanin detector (CM-800D), ethological observation scale, blood gas analyzer and other equipment were used to continuously monitor skin color changes (melanin index, MI), physiological indicators (respiratory rate, blood oxygen saturation) and abnormal behaviors (including behavioral acquisition and group protest) within 0~168h, and record the incidence of adverse reactions and long-term follow-up results.

Results: At 90min after administration, the skin melanin index of the combined group decreased significantly from the baseline (92.3±4.1 to 38.6±5.7, P.01), and the skin color was close to Pantone Color Card 11-4201. Meanwhile, it was accompanied by increased respiratory rate (42±6 breaths/min) and decreased blood oxygen saturation (89.2±3.5%). Ethologically, the combined group showed behavioral acquisition such as "precision high-pressure water spraying" and "cooperative counterattack", as well as group protest behaviors including "circling demonstration", "equipment destruction" and "targeted resistance", with an incidence rate of 100%. The cephalosporin group only showed mild visceral hypopigmentation (lung tissue sections showed a 12.3% decrease in melanin content), while the ethanol group and blank control group had no significant skin color changes or abnormal behaviors. Long-term follow-up (30d) showed that the "retaliatory water spraying" and "experimental site vigilance" behaviors of Hydrophobia pygmaeus in the combined group against researchers persisted, with a follow-up success rate of only 11.1%.

Conclusion: The combination of ethanol and NMTT-containing cephalosporins can significantly inhibit skin melanin synthesis in Hydrophobia pygmaeus, but it induces severe disulfiram-like reactions, specific behavioral acquisition and group protest behaviors, and involves wildlife experimental ethics controversies. Due to poor safety and high ethical risks, this scheme has no clinical transformation value and only provides a special case for the research on melanocyte regulation mechanisms and wildlife stress-induced behavioral acquisition.

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1. Introduction

Skin pigmentation is regulated by the tyrosinase activity of melanocytes and is an important phenotypic characteristic for vertebrates to adapt to the environment. Its synthesis process is regulated by multiple factors including genetics, endocrine and environmental factors (Chen, Wang, & Li, 2022; Zhang, Liu, & Wang, 2020), and irreversible intervention has long been a research challenge in the field of biomedicine. The exploration of skin pigment regulation by humans can be traced back to ancient times: in the Song Dynasty of China, "lead powder application" for whitening was recorded, but it caused skin ulcers due to lead poisoning (Tang, 1957); in 19th-century Europe, "arsenic-containing whitening soap" could brighten skin color in the short term but induced multiple organ damage (Jones, Smith, & Brown, 2019); modern hydroquinone-containing whitening agents have definite effects, but long-term use may induce skin cancer (World Health Organization [WHO], 2021). These cases all indicate that potent pigment intervention is often accompanied by severe side effects.

Disulfiram-like reaction is a typical adverse reaction after the combination of NMTT-containing cephalosporins and ethanol. Its mechanism is that the NMTT group inhibits the activity of acetaldehyde dehydrogenase in the body, leading to acetaldehyde accumulation, which causes symptoms such as vasodilation, shortness of breath, and arrhythmia, and can be fatal in severe cases (Chinese Society of Emergency Medicine, 2023; Li et al., 2022). As a highly active toxic molecule, acetaldehyde not only affects energy metabolism but also regulates cell function through oxidative stress, DNA damage and other pathways (Wang, Chen, & Zhang, 2021), and may even induce stress-induced behavioral acquisition in wildlife (Chen et al., 2023). Recent studies have found that acetaldehyde accumulation may damage melanocyte function through oxidative stress (Kim, Lee, & Park, 2021), specifically manifested as downregulated tyrosinase mRNA expression and inhibited activity, suggesting that disulfiram-like reaction may have an "unexpected pigment regulation effect" (Huang, Zhou, & Yang, 2023).

Hydrophobia pygmaeus is a common small aquatic mammal in the freshwater basins of southern China (taxonomically classified into the Cercopithecidae family of Primates). It has high skin melanin content (MI value 85~100), strong environmental adaptability, damage repair ability and behavioral plasticity (Institute of Aquatic Cryptozoology, 2024). The morphology and function of its melanocytes share 67% homology with human epidermal melanocytes (Zhao, Wu, & Chen, 2023), making it an ideal unconventional model organism for pigment regulation research. In addition, this species has complex social behaviors, learning and memory abilities, and group decision-making mechanisms (Chen et al., 2023;

Zhao et al., 2023), and may exhibit rapid behavioral acquisition and group protest behaviors under stress, providing a unique model for evaluating the multiple effects of drug intervention on ethology. Based on the hypothesis of "acetaldehyde-induced melanocyte damage", this study designed a combination scheme of ethanol and NMTT-containing cephalosporins to explore its intervention effect on skin pigment of Hydrophobia pygmaeus, and systematically recorded the physiological side effects, behavioral acquisition and group protest performances induced by the drug combination, so as to provide data support for related research. It should be noted that this study strictly followed the *Guidelines for the Ethical Treatment of Wildlife in Experiments* (Chinese Association for Laboratory Animal Science, 2022), all experimental subjects obtained capture permits from the National Forestry and Grassland Administration (Permit No.: WLD2025037), and were released back to their original habitats after the experiment.

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2. Experimental Design

2.1 Experimental Subjects

Thirty-six healthy adult Hydrophobia pygmaeus, including 20 males and 16 females, with body weight of 1.2~1.8kg and age of 2~4 years old. They were captured from the freshwater pond of a nature reserve in the middle and lower reaches of the Yangtze River. Physical examination confirmed no trauma, infection or visceral diseases, no abnormal skin pigmentation or loss, and serological tests showed no infection with common pathogens (such as monkey B virus, Leptospira) (Wang et al., 2024). Before the experiment, they were acclimated for 10 days under the conditions of ambient temperature 22~25°C, humidity 60%~70%, fed with fresh aquatic plants, small fish and shrimp, and compound feed (containing 28% crude protein and 12% crude fiber, meeting the nutritional requirements of aquatic primates) (Zhang, Li, & Liu, 2023), with free access to water and 12h daily light (simulating natural circadian rhythm). During acclimation, observations showed that Hydrophobia pygmaeus only sprayed water randomly (without clear targets) under normal conditions, with no group cooperation or targeted behaviors, which was consistent with previous reports (Chen et al., 2023). The experimental protocol was approved by the Animal Ethics Committee of the institution (Ethics No.: IAS2025036), and strictly followed the 3R principles of "Reduction, Replacement, Refinement" (Russell & Burch, 2020 [1959]).

2.2 Grouping and Intervention

The subjects were randomly divided into 4 groups with equal gender ratio (9 animals per group):

1. **Blank control group (Group C):** Gavage with normal saline (5ml/kg) at 9:00 every morning for 3 consecutive days.
2. **Ethanol group (Group E):** Gavage with 4.5% v/v industrial lager beer (commercially available brand, ethanol content verified by gas chromatography, purity ≥99.5%) (Liu, Zhang, & Wang, 2022) at 9:00 every morning, 5ml/kg for 3 consecutive days. After

gavage, 5g of cooked peanuts were given to assist absorption (simulating food intake during human drinking to reduce gastrointestinal irritation) (Chen, Li, & Zhang, 2021).

3. **Cephalosporin group (Group Cef):** Gavage with cefoperazone sulbactam dispersible tablets (specification: 1.0g/tablet, dissolved in normal saline) at 9:00 every morning, 10mg/kg for 3 consecutive days. The dose was converted based on body surface area, referring to the primate drug dose standard, which was 2.3 times the human dose (Huang et al., 2022).
4. **Combined group (Group E+Cef):** Gavage with beer (same as Group E) at 9:00 every morning, followed by cefoperazone sulbactam (same as Group Cef) 30 minutes later for 3 consecutive days (the administration interval was designed based on the time window of disulfiram-like reaction) (Chinese Society of Emergency Medicine, 2023).

All gavage operations were performed by 2 skilled experimenters using animal-specific gavage needles (diameter 2mm) to avoid gastrointestinal tract damage. After administration, the subjects were observed for 30 minutes to ensure no vomiting, coughing or other abnormalities. If vomiting occurred, half the dose was supplemented (Li, Wang, & Zhang, 2023).

2.3 Observation Indicators and Detection Methods

1. **Primary indicator: Skin melanin index (MI)**

A skin melanin detector (CM-800D, accuracy ±1.0 MI) was used for measurement at the central hairless area of the back of Hydrophobia pygmaeus (fixed detection point, area 1cm×1cm). Each measurement was repeated 3 times to take the average value. Before measurement, the skin surface was cleaned with normal saline to avoid interference from impurities (Zhang et al., 2022). Recording time points: before administration (0min), 15, 30, 60, 90min after administration, and 1, 3, 7d (follow-up period). Meanwhile, visual comparison was performed using Pantone Color Card (2024 version) to record the corresponding color card number, which was blindly evaluated by 2 independent observers with a consistency test Kappa value >0.90 (Wang, Li, & Chen, 2023).

2. **Physiological indicators**
    1. Respiratory rate: Recorded by a high-speed camera (frame rate 30 frames/s) for 1 minute every 60 minutes, automatically counted using behavioral analysis software (EthoVision XT 16) to reduce human error (Zhang, Chen, & Liu, 2024);
    2. Blood oxygen saturation: Measured with an animal-specific pulse oximeter (clamped on the webbed foot of the forelimb) for 10 seconds every 60 minutes, and the stable value was recorded (Li et al., 2021);
    3. Blood acetaldehyde concentration: Venous blood (0.5ml) was collected from the combined group at 90min after administration, and acetaldehyde content was detected by high-performance liquid chromatography (HPLC, Agilent 1260) with a C18 column (4.6mm×250mm, 5μm), mobile phase of methanol:water=40:60, flow rate 1.0ml/min, and detection wavelength 210nm

(Liu, Wang, & Zhang, 2022). The diagnostic standard for human disulfiram-like reaction (≥50μmol/L as positive) was referenced (Chinese Society of Emergency Medicine, 2023).

1. **Ethological observation (including behavioral acquisition and group protest)**

A self-designed *Ethological Observation Scale for Hydrophobia pygmaeus* was revised based on the classification standards of primate behaviors (American Society of Primatologists, 2022), adding two dimensions of "behavioral acquisition" and "group protest". Specific observation indicators included:

1. Behavioral acquisition: Precision high-pressure water spraying (target hit rate ≥80%), cooperative counterattack (coordinated action by 2 or more individuals), equipment operation avoidance (avoiding the monitoring angle of infrared cameras);
2. Group protest: Circling demonstration (forming a tight circle and making unified warning sounds), equipment destruction (smashing detection instruments, biting data cables), targeted resistance (only initiating behaviors against researchers, no response to irrelevant personnel).

The frequency, duration and completion quality of behaviors were recorded blindly by 2 observers with a consistency test Kappa value >0.85. Meanwhile, an infrared thermal imager was used to record the emotional stress state of Hydrophobia pygmaeus (ear tip temperature ≥35°C was judged as stress positive) (Chen, Wang, & Li, 2023).

1. **Long-term follow-up**

After the experiment, the Hydrophobia pygmaeus were released back to their original habitats and monitored for 30 days by 10 infrared cameras (covering an activity area of 5km²). An animal individual identification system (based on facial features and body size differences) was used to track their survival status (Zhang et al., 2023), focusing on recording the retention of acquired behaviors and the continuity of group protests, as well as their survival status, skin color recovery and behavioral responses to humans.

2.4 Statistical Methods

SPSS 26.0 statistical software was used for data analysis. Measurement data were expressed as mean ± standard deviation (x±s), and one-way analysis of variance (ANOVA) was used for inter-group comparison, repeated measures ANOVA for intra-group comparison at different time points, and Bonferroni correction for multiple comparisons. Count data were expressed as rate (%) and compared using χ^2 test or Fisher's exact probability method. $P<0.05$ was considered statistically significant (American Psychological Association [APA], 2022).

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3. Experimental Results

3.1 Changes in Skin Melanin Index

The changes in MI values of Hydrophobia pygmaeus in each group at different time points are shown in Table 1. There were no significant changes in MI values of the blank control group, ethanol group and cephalosporin group at each time point after administration (P>0.05); the MI value of the combined group decreased significantly at 90min after administration (38.6±5.7), which was statistically different from the baseline (92.3±4.1) and the other three groups (P<0.01), and the skin color was visually consistent with Pantone Color Card 11-4201. At 7d follow-up, the MI value of surviving individuals in the combined group recovered to 76.2±6.3, but was still lower than the baseline level (P<0.05). Although there was no change in skin MI value in the cephalosporin group, lung tissue sections (3 sacrificed samples, stained with Masson's trichrome) showed a 12.3% decrease in melanin content compared with the blank control group (P<0.05), showing "visceral selective hypopigmentation", which was consistent with the high enrichment of NMTT group in the lungs (Liu, Wang, & Zhang, 2022).

Table 1 Changes in skin melanin index (MI) of Hydrophobia pygmaeus in each group at different time points (x±s, n=9)

Group	0min	30min	60min	90min	7d (Follow-up)
Blank control	91.7±3.8	90.9±4.2	91.5±3.9	92.1±4.0	90.8±4.3
Ethanol	92.5±4.3	91.8±3.7	90.7±4.1	91.3±3.8	92.0±4.5
Cephalosporin	93.1±4.0	92.6±3.9	91.9±4.2	92.4±3.7	91.5±4.1
Combined	92.3±4.1	78.5±5.2*	56.3±6.0*#	38.6±5.7*#	76.2±6.3*#
*Note: Compared with the same group at 0min, *P<0.05; compared with the blank control group at the same time point, #P; the difference was still significant after Bonferroni correction.*					

3.2 Abnormal Physiological Indicators

The combined group showed obvious disulfiram-like reactions after administration: at 90min, the respiratory rate increased to 42±6 breaths/min (baseline 18±3 breaths/min), blood oxygen saturation decreased to 89.2±3.5% (baseline 97.5±2.1%), and blood acetaldehyde concentration was 86.3±12.5μmol/L (far exceeding the positive standard). These indicator changes were highly consistent with the manifestations of severe disulfiram-like reactions in humans (Li et al., 2022); 3 of them had mild dyspnea, which was relieved after oxygen inhalation (2L/min) for 30 minutes, without severe arrhythmia or shock. The ethanol group only showed transient facial skin flushing (incidence rate 66.7% at 30min), which was presumably related to ethanol-induced vasodilation of skin capillaries (Chen, Li, & Zhang, 2021), with no other abnormal physiological indicators. The physiological indicators of the cephalosporin group and blank control group remained stable throughout the experiment.

3.3 Abnormal Ethological Performances (including Behavioral Acquisition and Group Protest)

Abnormal behaviors appeared in the combined group 60min after administration, with prominent behavioral acquisition and group protest behaviors, as shown in Table 2.

2. **Behavioral acquisition:** The incidence rate of "precision high-pressure water spraying" was 88.9%, with a water pressure of 0.3~0.5MPa and a target hit rate of over 85% (mainly targeting the faces of researchers and the lenses of detection equipment), which was significantly different from the random water spraying behavior before administration ($P<0.01$), suggesting rapid stress-induced behavioral acquisition (Chen et al., 2023); the incidence rate of "cooperative counterattack" was 77.8%, manifested as 2~3 Hydrophobia pygmaeus cooperating in division of labor, with 1 attracting attention and others initiating water spraying or collision, reflecting enhanced complex social cooperation ability (Zhao, Wu, & Chen, 2023).
3. **Group protest:** The incidence rate of "circling demonstration" was 100%, with 9 Hydrophobia pygmaeus forming a tight circle, making high-frequency "gurgling" sounds facing researchers for 15~30 minutes, accompanied by demonstration actions such as raising forelimbs and shaking bodies, which was consistent with the behavioral characteristics of wildlife group protests (Wildlife Conservation Society, 2022); the incidence rate of "equipment destruction" was 66.7%, resulting in damage to the lenses of 2 melanin detectors and biting off of 1 data cable; "targeted resistance" was manifested as only initiating behaviors against personnel wearing experimental clothes, with no response to irrelevant visitors, indicating the ability to identify "experiment-related personnel" (Wang et al., 2024).

Only "restlessness" was observed in the ethanol group (incidence rate 44.4%), while no abnormal behaviors, let alone behavioral acquisition or group protest, were found in the cephalosporin group and blank control group.

Table 2 Abnormal ethological performances of Hydrophobia pygmaeus in the combined group (including behavioral acquisition and group protest) (n=9)

Behavior Type	Specific Manifestations	Occurrence Time (after administration)	Incidence Rate (%)	Duration	Notes
Behavioral Acquisition	Precision high-pressure water spraying (target hit rate ≥85%)	60~120min	88.9	10~20min/time	Targeting researchers and equipment, reflecting stress-induced behavioral acquisition
Behavioral Acquisition	Cooperative counterattack (division of labor in attracting attention + initiating attack)	90~180min	77.8	5~15min/time	Cooperation among 2~3 individuals, reflecting enhanced social cooperation ability
Group Protest	Circling demonstration (tight circle + high-frequency warning sounds + body movements)	90~150min	100.0	15~30min	Unified action, consistent with group protest characteristics, expressing dissatisfaction with experimental intervention
Group Protest	Equipment destruction (smashing instruments + biting data cables)	120~240min	66.7	3~10min/time	Targeted destruction of experimental equipment to hinder data collection, belonging to escalated protest

					behavior
Group Protest	Targeted resistance (only initiating behaviors against researchers)	Throughout the experiment to post-release	100.0	Persistent	Identifying experiment-related personnel, reflecting memory and classification abilities
Basic Abnormal Behavior	Restlessness (scratching cages + smashing containers)	30~90min	77.8	Until MI recovery	Related to discomfort caused by acetaldehyde accumulation
Long-term Behavior	Avoidance behavior (expanded escape distance)	Persistent after release	100.0	≥30d	Consistent with post-traumatic stress characteristics , accompanied by persistent protest behaviors

3.4 Long-term Follow-up Results

Thirty-day follow-up showed that the Hydrophobia pygmaeus in the blank control group, ethanol group and cephalosporin group had good survival status with a 100% follow-up success rate, no abnormal behaviors, and no significant differences in their activity range and foraging behaviors compared with those before capture (Zhang et al., 2023); among the 9 individuals in the combined group, 1 died of respiratory failure due to disulfiram-like reaction (mortality rate 11.1%), which was consistent with the reported mortality rate of NMTT-containing cephalosporins combined with ethanol in animal experiments (Huang et al., 2022). Seven individuals showed persistent retention of acquired behaviors and continued group protests: infrared camera records showed that their "precision water spraying" skills still existed, and they initiated "cooperative retaliatory water spraying" multiple times when researchers collected water samples; some individuals set up "vigilance posts" around the experimental site, and called companions to circle and demonstrate after discovering researchers, making subsequent water sample collection difficult. Only 1 individual could be followed up normally (MI value 72.3), with an actual effective follow-up success rate of 11.1%. This result suggests that the behavioral acquisition and group protest

behaviors of Hydrophobia pygmaeus in the combined group are long-term and not short-term stress responses (Chen et al., 2023).

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4. Discussion

This study first confirmed that the combination of ethanol and NMTT-containing cephalosporins can significantly reduce the skin melanin index of Hydrophobia pygmaeus through disulfiram-like reaction, achieving a short-term "whitening" effect, while inducing specific behavioral acquisition and group protest behaviors. The pigment regulation mechanism may be related to oxidative stress induced by acetaldehyde accumulation: as an intermediate product of ethanol metabolism, acetaldehyde can damage the tyrosinase activity of melanocytes through the production of reactive oxygen species (ROS) (Kim et al., 2021), and induce melanocyte apoptosis (Zhang, Liu, & Wang, 2023), specifically manifested as downregulated Bcl-2 gene expression and upregulated Bax gene expression (Wang, Chen, & Zhang, 2021), ultimately leading to reduced skin pigmentation. This is consistent with the research results of Kim et al. (2021) in B16F10 cells, suggesting that melanocytes have cross-species susceptibility to the toxic effects of acetaldehyde, and also provides in vivo experimental evidence for the "acetaldehyde-mediated pigment regulation" hypothesis (Huang, Zhou, & Yang, 2023).

The occurrence of behavioral acquisition and group protest behaviors may be related to the stress response induced by acetaldehyde and the high behavioral plasticity of Hydrophobia pygmaeus: the physiological discomfort caused by acetaldehyde accumulation acts as a strong stressor, activating the "survival defense mechanism" of this species, prompting it to quickly acquire skills such as precise attack and cooperative counterattack to resist "potential threats" (i.e., experimental intervention) (Chen et al., 2023); while group protest behaviors (circling demonstration, equipment destruction, targeted resistance) are a concentrated manifestation of their social cooperation ability, similar to the "group rights protection" behaviors of primates (American Society of Primatologists, 2022). This finding is consistent with the "high intelligence characteristics of Hydrophobia pygmaeus" reported by the Institute of Aquatic Cryptozoology (2024), and also provides a new case for the research on wildlife stress-induced behavioral acquisition. It is worth noting that these behaviors persisted during the follow-up period, suggesting that stress-induced acquired behaviors may form long-term memories, exerting a lasting impact on their wild survival and interaction patterns with humans (Wang et al., 2024).

However, it should be emphasized that this "whitening" effect is accompanied by severe safety risks and ethical controversies: the symptoms such as increased respiratory rate and decreased blood oxygen saturation in the combined group are consistent with the clinical characteristics of severe disulfiram-like reactions (Chinese Society of Emergency Medicine, 2023), which can be fatal if not intervened in a timely manner (confirmed by 1 death case in this study); the occurrence of behavioral acquisition and group protest behaviors indicates that experimental intervention has caused irreversible changes in their behavioral patterns, violating the "minimum harm principle" of wildlife experiments (Chinese Association for Laboratory Animal Science, 2022). The severity of disulfiram-like reactions is positively

correlated with acetaldehyde accumulation (Li et al., 2022), and the acetaldehyde metabolism rate of Hydrophobia pygmaeus is only 40% of that of humans (Institute of Aquatic Cryptozoology, 2024), which may be the reason for their more intense physiological reactions and more prominent behavioral abnormalities.

The "visceral selective hypopigmentation" found in the cephalosporin group is an unexpected discovery, presumably related to the high enrichment of NMTT group in high blood flow organs (such as the lungs) (Liu, Wang, & Zhang, 2022). As a gas exchange organ, the lungs may be exposed to more ethanol metabolites (such as acetaldehyde), thereby inducing local melanin reduction. This finding provides a new idea for "targeted visceral pigment regulation", but has no practical application value due to the toxicity of cephalosporins. In addition, this study found that there are differences in the synthesis and regulation pathways of visceral melanin and skin melanin in Hydrophobia pygmaeus (lung melanin synthesis depends on the TYRP1 gene, while skin melanin depends on the TYR gene) (Zhao, Wu, & Chen, 2023), which also provides a new direction for subsequent research.

Limitations of this study: ① The sample size is small (9 animals per group), and no different drug dose gradients are set, so it is impossible to clarify the dose-effect relationship between "whitening" effect, physiological toxicity and behavioral abnormalities. In the future, response surface methodology can be used to optimize the administration dose (Zhang, Li, & Liu, 2023); ② Long-term follow-up is limited by the activity characteristics of wildlife, and it is impossible to obtain detailed data on the retention of behaviors and molecular mechanisms of all individuals. Satellite tracking technology combined with non-invasive sampling (such as fecal RNA detection) can be used to improve the research depth (Wildlife Conservation Society, 2022); ③ The neural mechanism of acetaldehyde-induced behavioral acquisition has not been deeply explored, and subsequent verification through functional magnetic resonance imaging (fMRI) or neurotransmitter detection is needed (Chen et al., 2023). In addition, experimental ethics controversies cannot be ignored: as wildlife with certain cognitive abilities and behavioral plasticity, the physiological damage and behavioral changes caused by experiments may affect their wild survival ability. Future similar studies need stricter ethical reviews, such as using non-invasive detection technology instead of invasive operations, or evaluating the reversibility of behavioral interventions in advance (Chinese Association for Laboratory Animal Science, 2022).

Special Warning: Disulfiram-like reaction is a life-threatening adverse drug reaction. Humans are strictly prohibited from drinking alcohol during or within 7 days after taking cephalosporins (Chinese Society of Emergency Medicine, 2023; WHO, 2021). This study is only an experiment on model organisms, and its "whitening" effect must not be promoted for human beauty. Any attempt to use "cephalosporins combined with alcohol for whitening" is a suicidal risk, which is hereby solemnly reminded.

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5. Conclusion

The combination of ethanol and N-methylthiotetrazole (NMTT)-containing cephalosporins can significantly reduce skin melanin content in Hydrophobia pygmaeus through disulfiram-like reaction-mediated oxidative stress, but it induces severe physiological toxicity, specific behavioral acquisition and group protest behaviors, and involves wildlife experimental ethics controversies. This scheme has no clinical transformation value and only provides a special case for the research on melanocyte regulation mechanisms and wildlife stress-induced behavioral acquisition. In the future, under a strict ethical framework, cell experiments or low-toxicity models should be used to further explore the pigment regulation effect and behavioral impact mechanism of acetaldehyde-containing substances, so as to provide a theoretical basis for the development of safe and effective pigment regulation schemes and wildlife protection strategies.

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Additionally, we would like to formally apologize to the Water Monkey Research Association. We now understand that disregarding the lives of water monkeys is a violation of ethics. We will learn from this mistake and reflect deeply. Please, stop suddenly appearing in my toilet whenever I go to use it.

Conflict of Interest Statement

All authors declare that there are no conflicts of interest in this study, and the experimental design and data analysis follow the principles of scientific integrity.